Global Olympic Sports Performance Analysis

Project Documentation — End-to-End Development Procedure

1. Project Overview

Global Olympic Sports Performance Analysis is a data analytics and visualization project that studies Olympic athlete-event and medal records. The project uses Tableau to identify patterns in countries, sports, medals, Olympic seasons, gender participation, and host cities. A Flask-based website was developed to present the Tableau dashboard and story through a professional web interface.

2. Project Objectives

- Analyze Olympic medal performance across countries.
- Identify sports producing high numbers of medals.
- Analyze gold, silver, and bronze medal distribution.
- Study medal trends across Olympic years.
- Compare Summer and Winter Olympic performance.
- Analyze medal-bearing participation by gender.
- Identify top Olympic cities by medal count.
- Create an interactive Tableau dashboard and an 8-scene Tableau story.
- Integrate the published Tableau views into a Flask website.

3. Dataset Collection

The project uses the 120 Years of Olympic History: Athletes and Results dataset. Two CSV files were used: athlete_events.csv and noc_regions.csv. The athlete dataset contains approximately 271,000 athlete-event records and fields including ID, Name, Sex, Age, Height, Weight, Team, NOC, Games, Year, Season, City, Sport, Event, and Medal. The NOC mapping file was used to associate NOC codes with country or region names.

4. Tableau Data Connection and Preparation

The two CSV files were connected in Tableau using the common NOC field. The relationship used was athlete_events.csv.NOC = noc_regions.csv.NOC. The data was then prepared for visualization.

5. Calculated Fields

Calculated Field	Purpose
Medal Count	Returns 1 for a medal-bearing record and 0 when Medal is null.
Gold Medal Count	Counts Gold medal-bearing records.
Silver Medal Count	Counts Silver medal-bearing records.
Bronze Medal Count	Counts Bronze medal-bearing records.
Country	Uses Region when available; otherwise uses NOC.
Athlete Count	Counts distinct athlete IDs.
Medal Winner	Classifies records as Medal or No Medal.

6. Tableau Visualizations

1. Top 10 Countries by Medal Count
2. Medals by Sport
3. Medal Type Distribution
4. Medals Over the Years
5. Top 10 Countries by Gold Medals
6. Medals by Season
7. Medals by Gender
8. Top 10 Sports by Gold Medals
9. Top 10 Olympic Cities by Medal Count
10. Top 10 Countries by Season

7. Dashboard Development

All major visualizations were combined into a Tableau dashboard titled **Global Olympic Sports Performance Analysis**. KPI cards were added for Total Medals, Total Countries, and Total Athletes. Four interactive filters were configured: Year, Season, Sport, and Country. The filters were applied across worksheets using the same data source.

8. Tableau Story Development

An 8-scene Tableau story titled **Olympic Sports Performance Story** was created to present the analysis as a sequence of insights:

1. The Olympic Performance at a Glance
2. Olympic Medal Trends: A Historical Overview
3. Nations Leading the Olympic Medal Race
4. Sports Driving Olympic Medal Success
5. The Gold Medal Leaders
6. Summer vs Winter: Olympic Medal Landscape
7. The Changing Landscape of Olympic Participation
8. Sports Producing the Most Gold Medals

9. Tableau Publishing

The completed dashboard and story were published using Tableau Public. The published views were then used as embedded content in the Flask website.

10. Flask Website Development

A Flask web application was developed using Python. The application renders an HTML template and provides a professional interface containing the project overview, objectives, statistics, Tableau dashboard, Tableau story, technology section, and responsive navigation.

Project structure:

```
Global_Olympic_Sports_Analysis/
■■■ app.py
■■■ requirements.txt
```

```
■■■ README.md
■■■ templates/
■ ■■■ index.html
■■■ static/
```

11. Running the Flask Application

Install dependencies:

```
python -m pip install -r requirements.txt
```

Run the application:

```
python app.py
```

Open the local website:

```
http://127.0.0.1:5000
```

12. GitHub Repository

The Flask website source code, requirements file, README documentation, and this project documentation are maintained in a public GitHub repository. The repository provides the source files required to review the web integration and project development.

13. Project Outcome

The final solution combines data preparation, Tableau visualization, dashboard design, data storytelling, and Flask web integration. It provides an interactive way to explore Olympic performance by country, sport, year, season, gender, and city.

14. Limitations

The dataset used in this project contains Olympic records through the 2016 Rio Olympics, so it does not contain actual Olympic data after 2016. Also, medal counts calculated directly from athlete-event records may repeat a team-event medal across participating athletes; therefore these counts should be interpreted as medal-bearing athlete-event records rather than official unique medal-event totals.

15. Skills Demonstrated

- Data Preprocessing
- Data Analysis
- Tableau Business Intelligence
- Data Visualization
- Dashboard Design
- Data Storytelling
- Python Flask Web Development
- HTML, CSS and JavaScript